

ORTA DOĞU TEKNİK ÜNİVERSİTESİ
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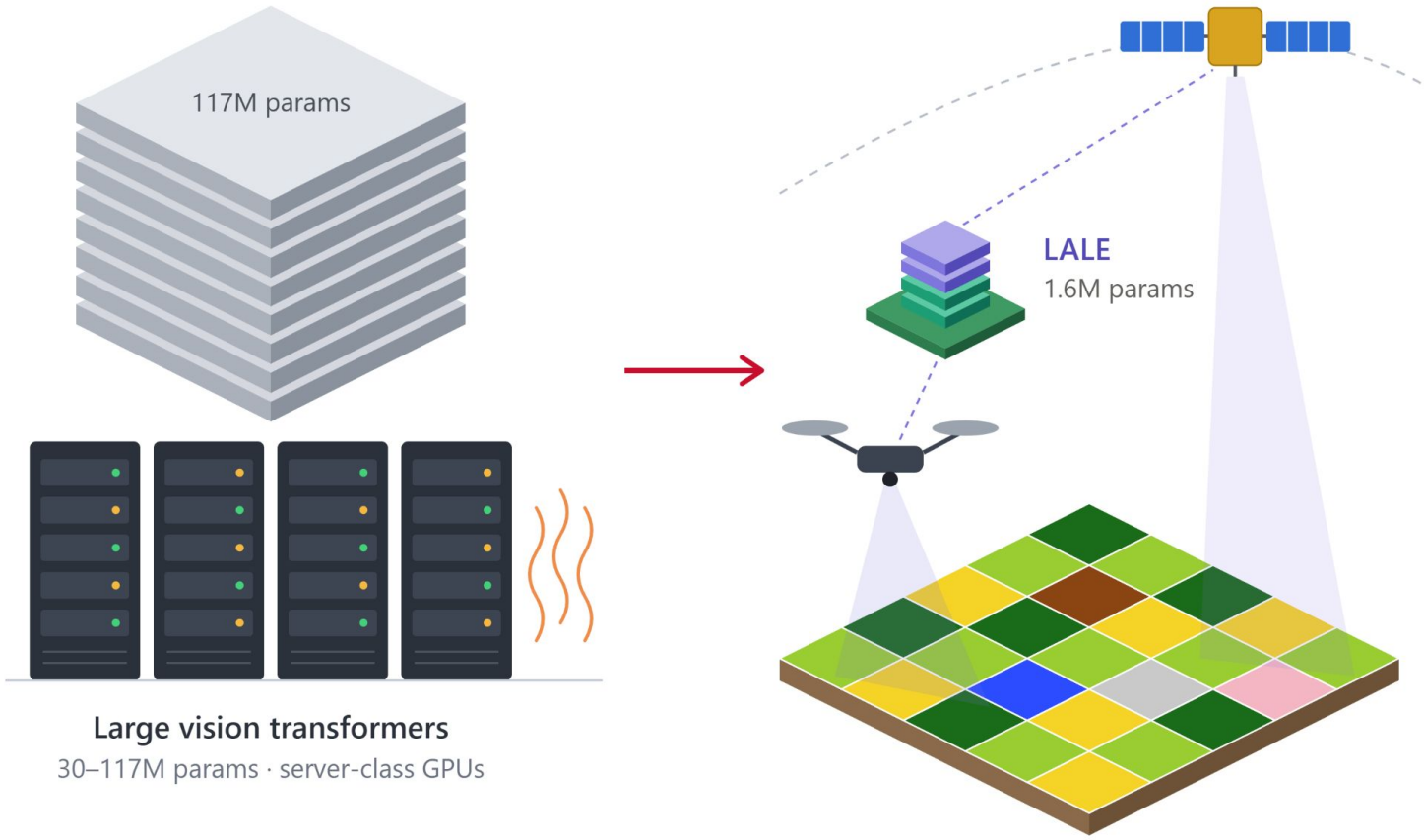
Lightweight-transformer Architecture for Land-cover Estimation (LALE)

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Lightweight-transformer Architecture for Land-cover Estimation (LALE)



117M → 1.6M
Parameters
up to 73× fewer

447 MB → 6 MB
Model size
up to 75× smaller

39.9 → 0.59
GMACs per image
up to 68× fewer

"Up to": DeiT3-Base (largest transformer baseline) vs. LALE-S1, ARAS400k

Abstract

- **Semantic segmentation** at scale requires models to balance **performance** with **efficiency**

- **L**ightweight-transformer **A**rchitecture for **L**and-cover **E**stimation **(LALE)**

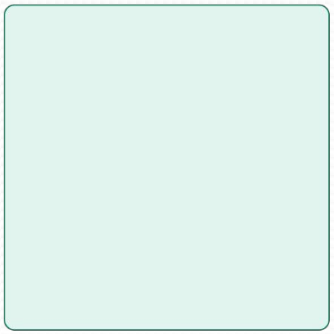
- Combines **convolutional** stages for **local** features with **transformer** stages for **global** context

- **Minimize** computational cost of **self-attention**

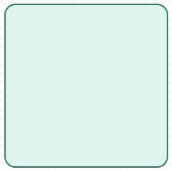
Introduction

- CNNs have **limited receptive** fields (local feature)
- ViTs cost **quadratic attention** (global context)
- Existing **hybrid** models inherit **heavy** Backbones
- Segmentation: balance **efficiency**-**performance**
- Use precise **local features** (convolution) and **global context** (attention)

Introduction



Stage 1
ConvMixer
64×64 · 32 ch
4,096 tokens



Stage 2
ConvMixer
32×32 · 64 ch
1,024 tokens

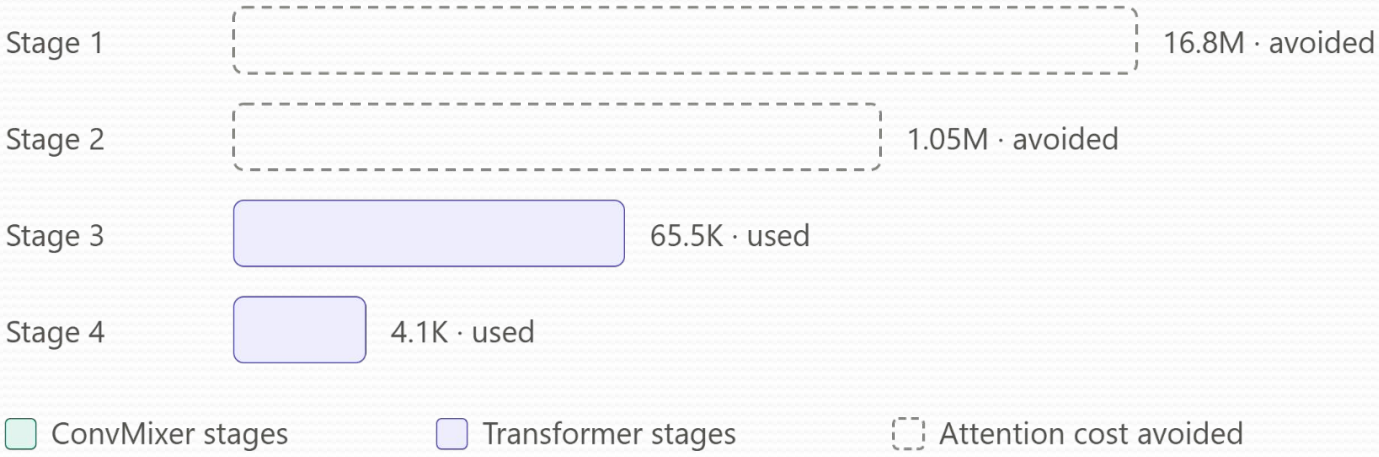


Stage 3
Transformer
16×16 · 128 ch
256 tokens



Stage 4
Transformer
8×8 · 256 ch
64 tokens

Self-attention matrix entries (N^2, log scale)



Literature

- Remote sensing challenges include extreme **scale variations**, **high resolutions**, and **heterogeneous backgrounds**

- **UNet** and **UNet++** established baseline spatial recovery using symmetric and densely cross-fused skip connections

- **Linknet**, **FPN**, and **PAN** introduced lightweight structures, multi-scale pyramids, and spatial attention mechanisms

Literature

- **DeepLabV3+** and **UPerNet**: advanced contextual mapping using atrous pooling modules and pyramid pooling

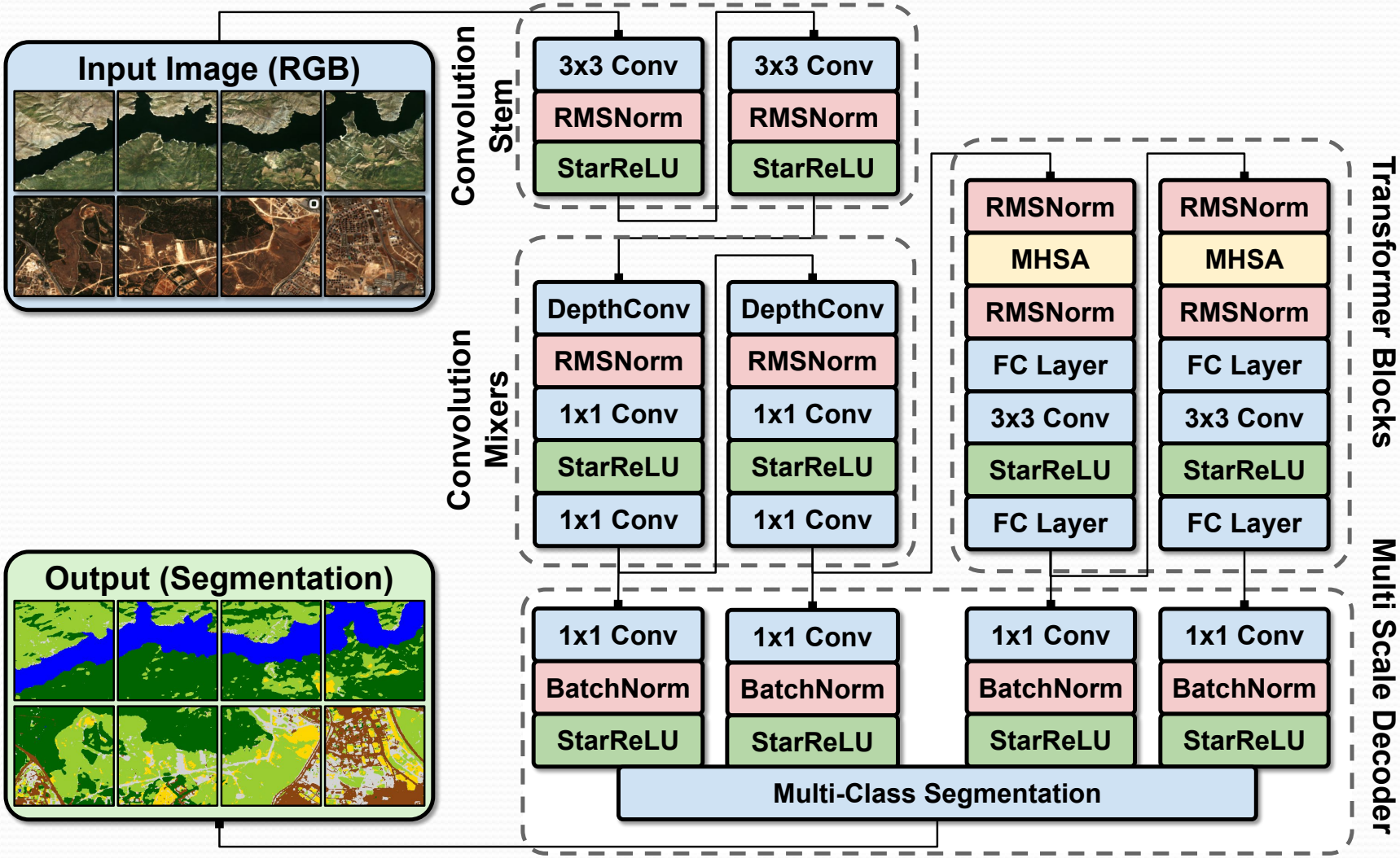
- **SegFormer**: positional-encoding-free transformer encoder for infinite receptive fields

- **DeiT3**: refined training recipes and advanced data augmentation boosting performance

Literature

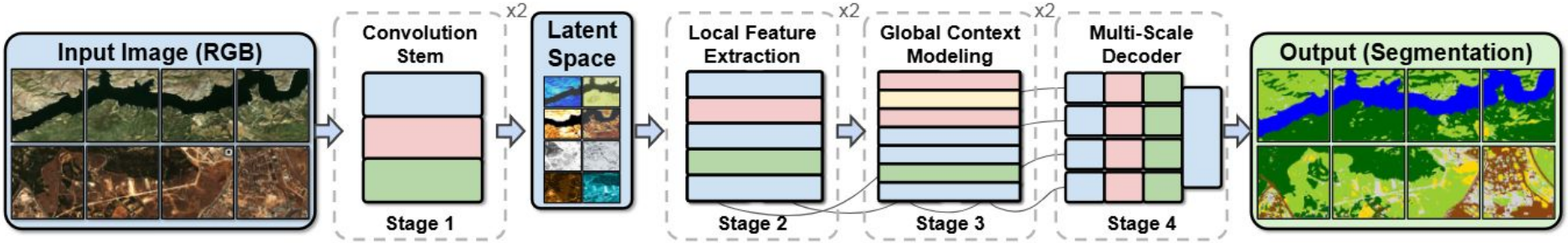
- **SwinV2**: post-normalization and log-spaced relative position biases
- **MaxViT**: multi-axis attention to combine local windows and global grids with linear complexity
- **EfficientFormer**: dimension-consistent design
- **FastViT**: structural reparameterization and RepMixer blocks minimize latency

Method


Input Image (RGB)
Output (Segmentation)
Convolution Stem
3x3 Conv
RMSNorm
StarReLU
3x3 Conv
RMSNorm
StarReLU
Convolution Mixers
DepthConv
RMSNorm
1x1 Conv
StarReLU
1x1 Conv
DepthConv
RMSNorm
1x1 Conv
StarReLU
1x1 Conv
Transformer Blocks
RMSNorm
MHSA
RMSNorm
FC Layer
3x3 Conv
StarReLU
FC Layer
RMSNorm
MHSA
RMSNorm
FC Layer
3x3 Conv
StarReLU
FC Layer
Multi Scale Decoder
1x1 Conv
BatchNorm
StarReLU
1x1 Conv
BatchNorm
StarReLU
1x1 Conv
BatchNorm
StarReLU
1x1 Conv
BatchNorm
StarReLU
Multi-Class Segmentation

Method

- Balances **global context** and **local detail** for resource-constrained remote sensing
- **ConvMixer** for high-resolution local features and transformers for low-resolution global context
- All-MLP decoder, **RMSNorm**, and **StarReLU** minimize computation and parameter count
- Comparable with **CNN**, **transformer**, and **hybrid** baselines on the **ARAS400k** and **LiTs** datasets
- Drastically **fewer parameters** and **GMACs** while maintaining high throughput and accuracy



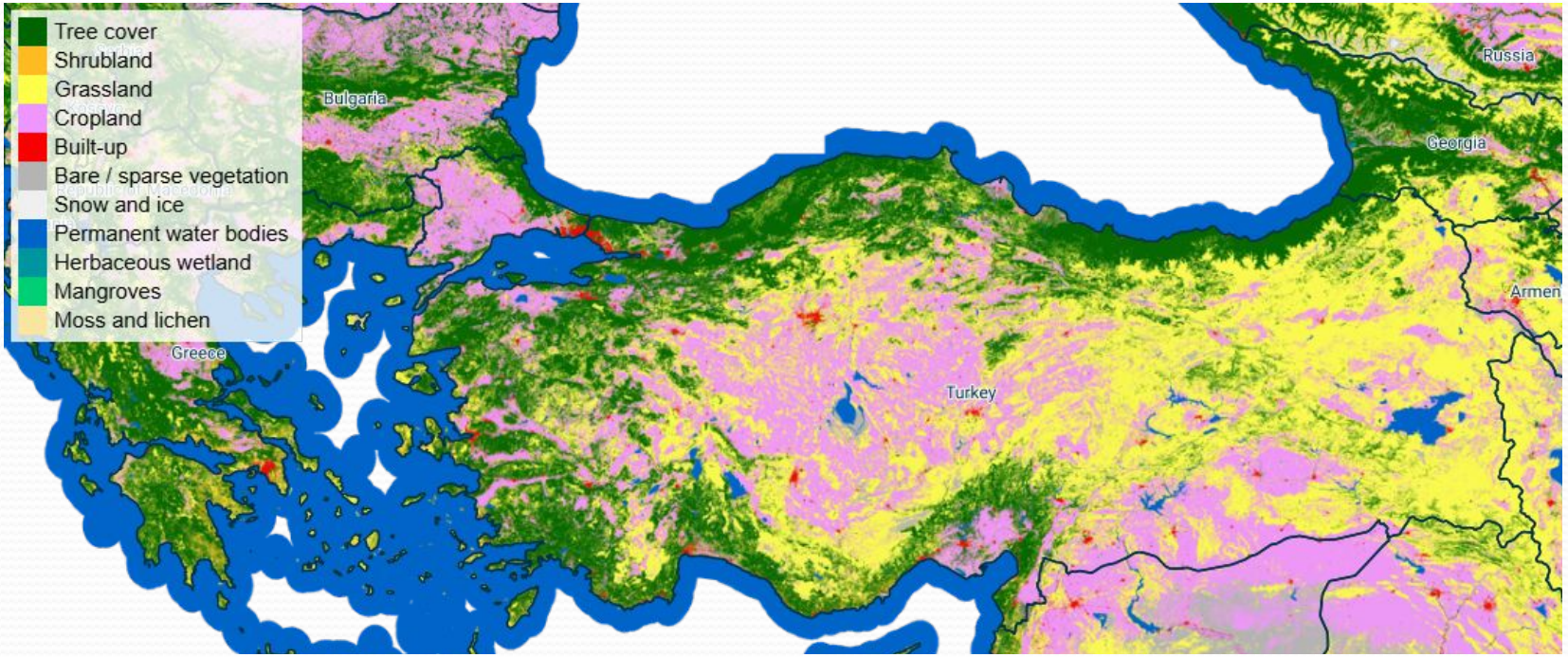
Dataset: ARAS400k

Diverse **ecosystems** and **biomes**

Sentinel-2 True color (RGB) and **ESA WorldCover**

Train: 80,192 + 300,000 synthetic **Val** and **Test**: 10,024

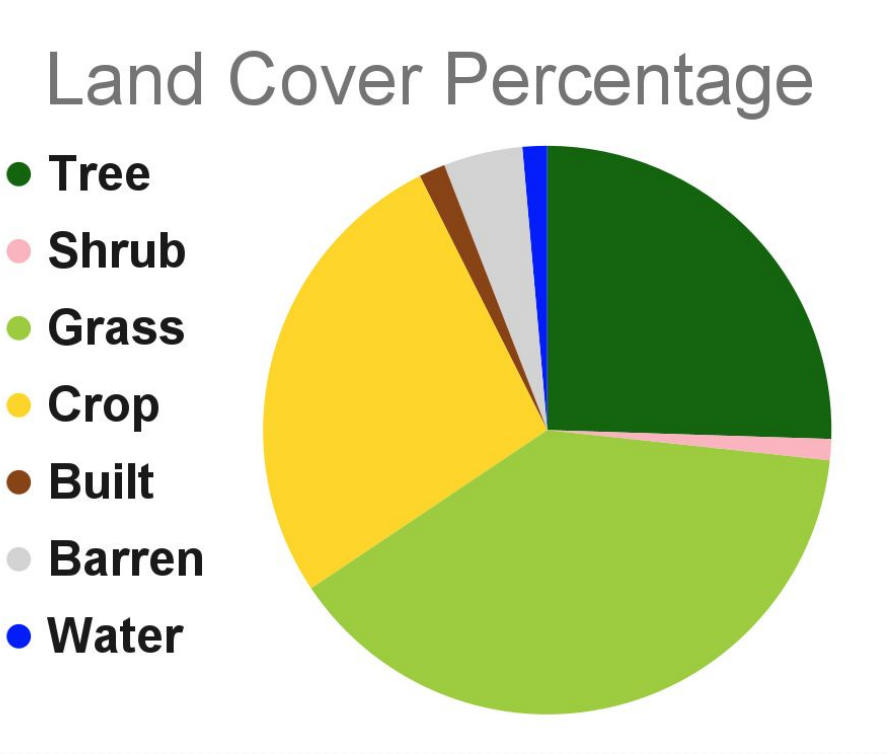
256 x 256 tiles geo-aligned RGB and Land-cover

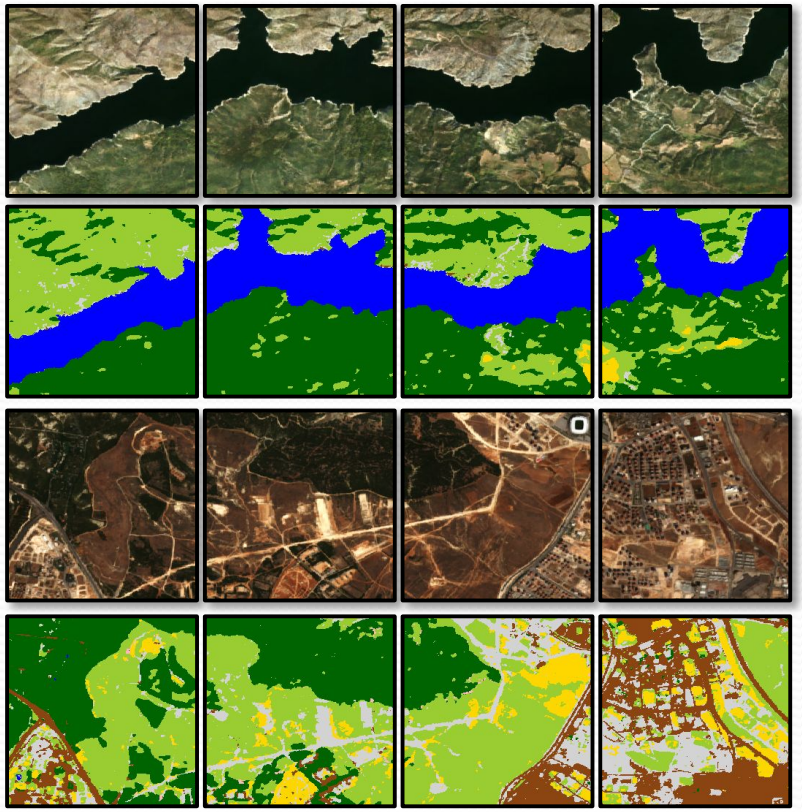


ESA WorldCover 11 class, ARAS400k merged into 7 classes

Dataset: ARAS400k

Percentage	Tree	Shrub	Grass	Crop	Built	Barren	Water
Cover	25.5	1.2	38.9	27.0	1.5	4.5	1.4
Presence	59.3	14.4	94.2	67.6	23.6	43.3	7.3
Color							

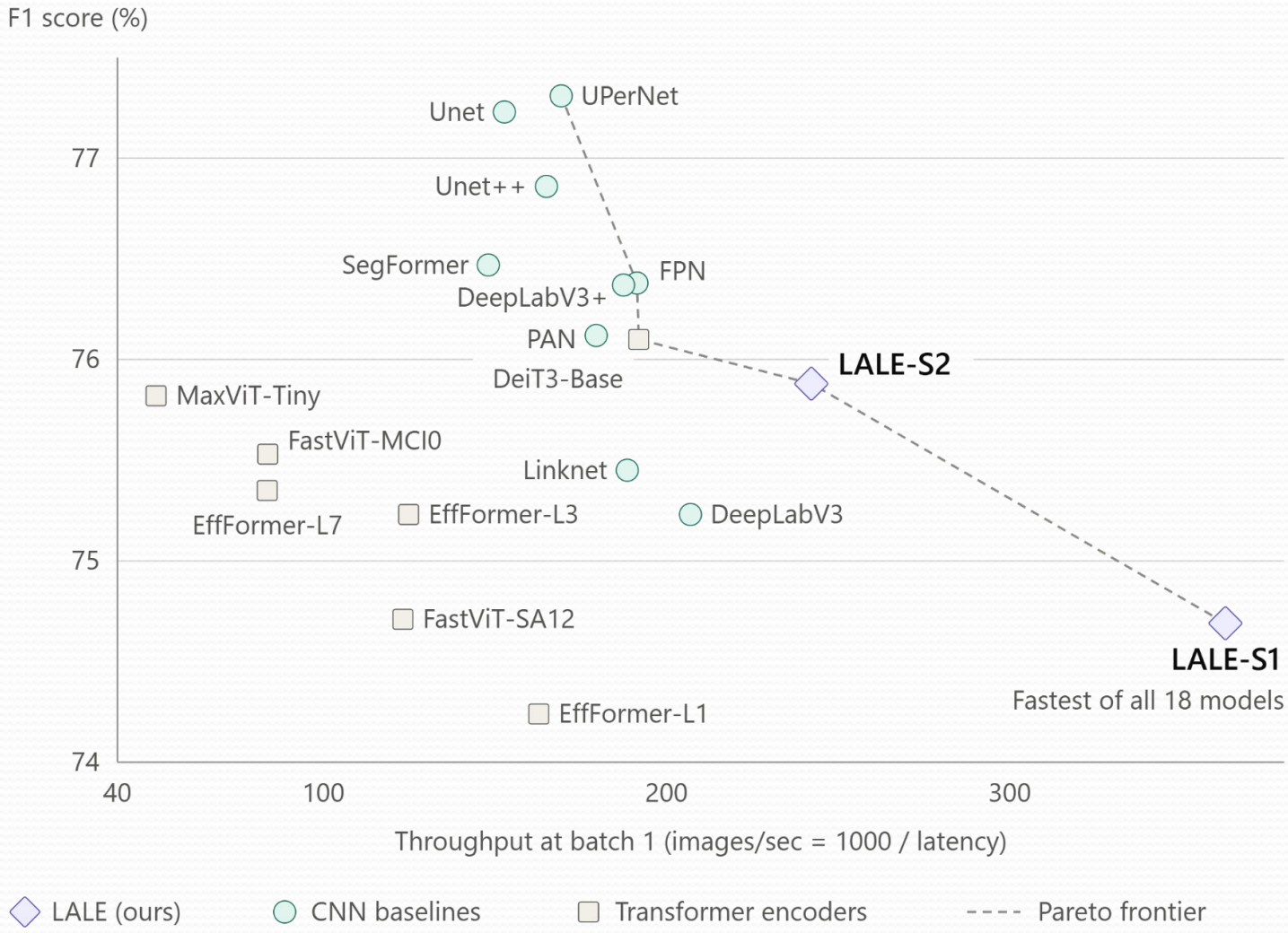




Results (ARAS)

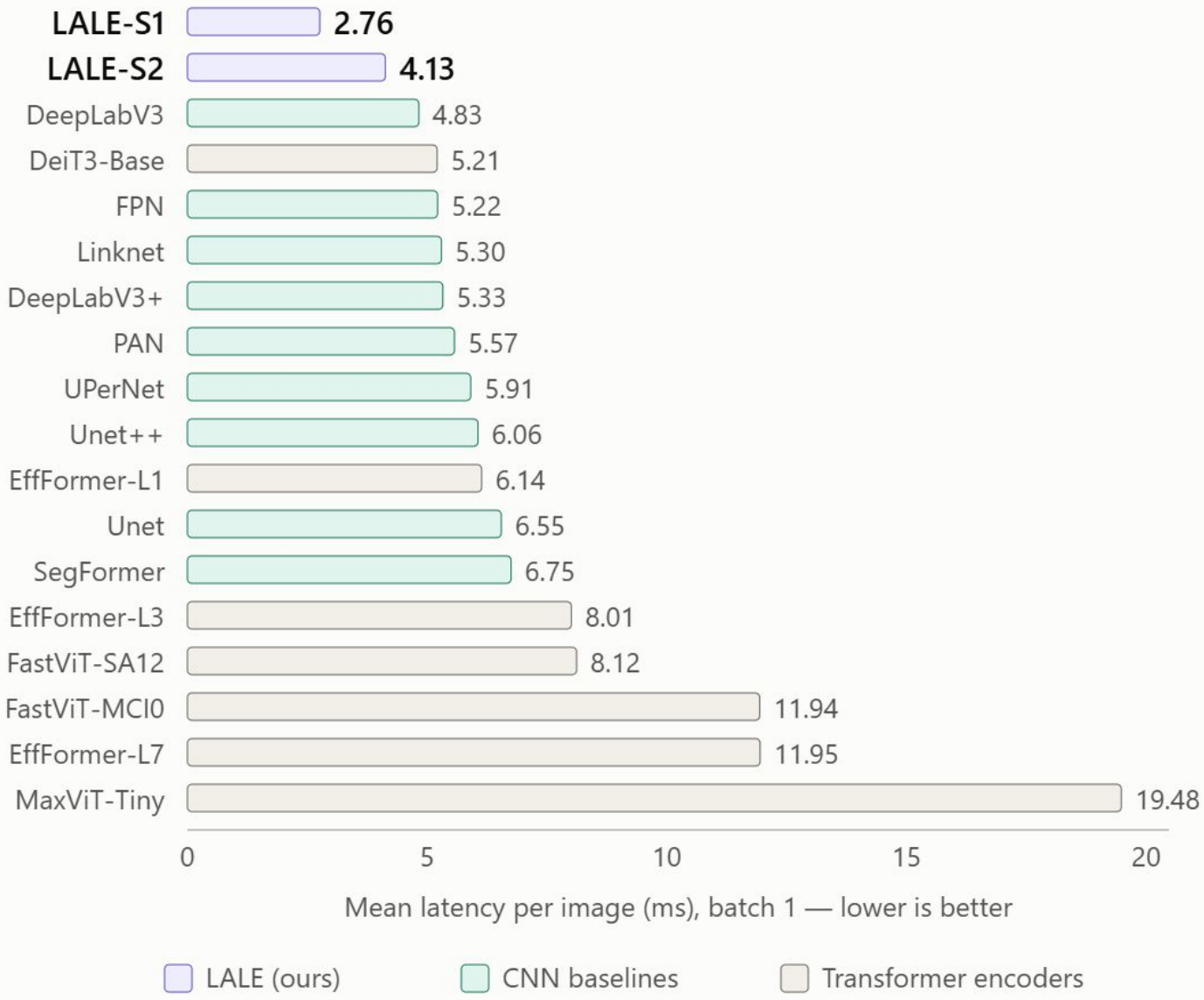
Architecture	F1	Accuracy	Precision	Recall	IoU	Size (MB)	Throughput (img/s, bs=1)	Params (M)	GMACs
DeepLabV3	75.23	84.22	76.00	74.58	63.01	28.04	207	7.3	6.44
DeepLabV3+	76.37	84.94	77.34	75.48	64.30	18.90	188	4.9	1.46
FPN	76.38	85.00	76.68	76.14	64.35	22.13	191	5.8	2.51
Linknet	75.45	84.09	76.21	74.76	63.21	16.06	189	4.2	0.58
PAN	76.12	84.57	76.18	76.14	63.93	15.80	180	4.1	0.98
Unet	77.23	85.09	77.13	77.53	65.28	24.02	153	6.3	3.05
UnetPlusPlus	76.86	85.00	77.17	76.60	64.90	25.24	165	6.6	5.62
UPerNet	77.31	85.53	77.83	76.84	65.42	44.49	169	11.6	13.62
Segformer	76.47	84.82	76.05	77.10	64.44	17.23	148	4.5	2.05
LALE-S1	74.69	83.16	75.18	74.26	62.25	5.98	363	1.6	0.59
LALE-S2	75.88	84.12	76.39	75.42	63.67	9.97	242	2.6	0.78
EffFormer-L1	74.24	83.13	73.33	75.64	61.81	113.97	163	29.8	23.17
EffFormer-L3	75.23	83.96	73.83	77.17	62.97	187.75	125	49.1	25.85
EffFormer-L7	75.35	84.00	74.36	76.89	63.13	383.22	84	100.3	32.16
DeiT3-Base	76.10	84.53	75.63	76.74	63.97	446.64	192	117.1	39.89
MaxViT-Tiny	75.82	84.46	75.55	76.24	63.64	232.26	51	60.8	33.13
FastViT-SA12	74.71	83.36	74.29	75.32	62.32	111.71	123	29.2	23.40
FastViT-MCI0	75.53	83.96	74.53	76.96	63.33	111.11	84	29.1	23.75

Results (ARAS)

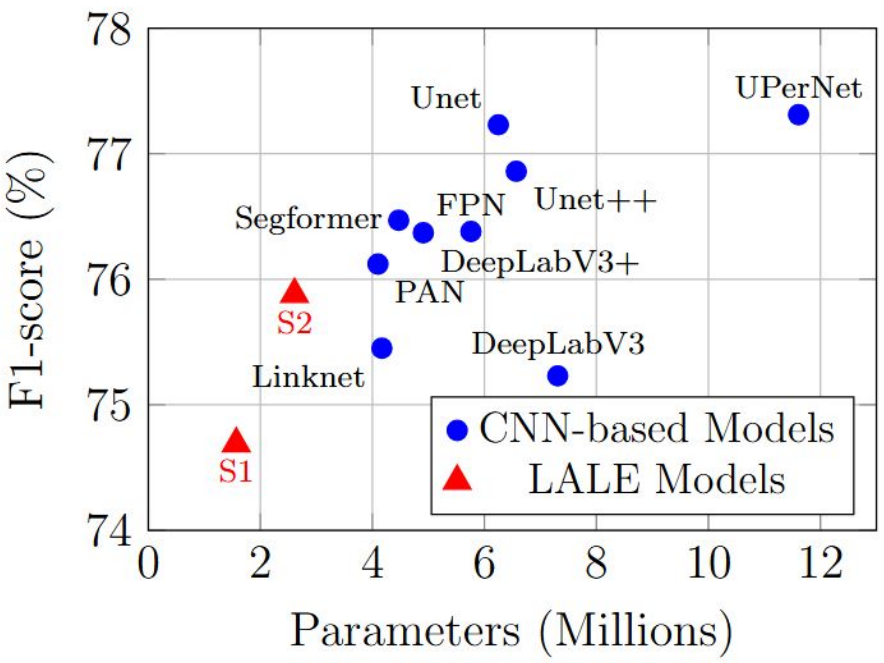


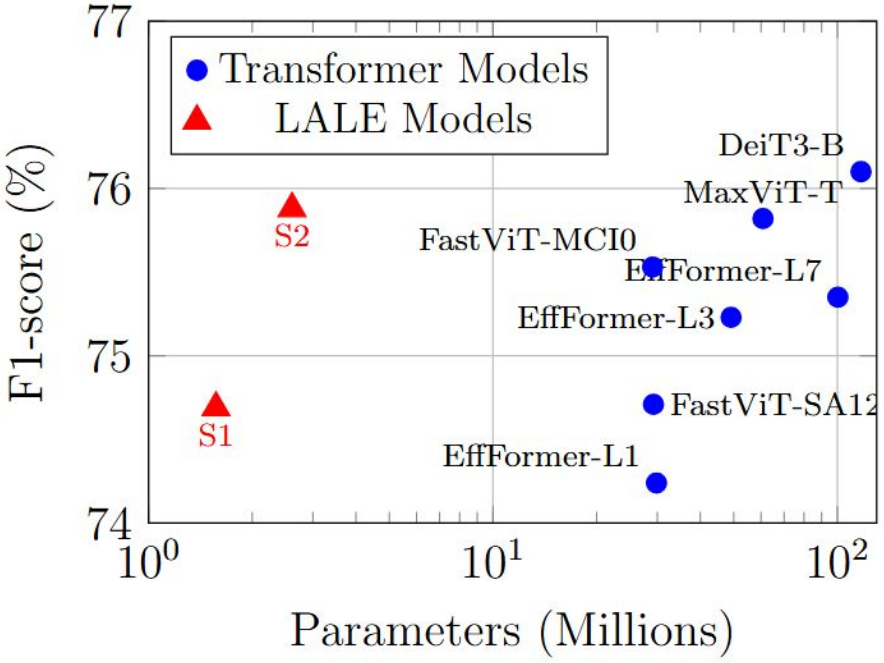
NVIDIA H100 80GB FP32, 256×256, mean of 100 runs after 10 warm-up run

Results (ARAS)

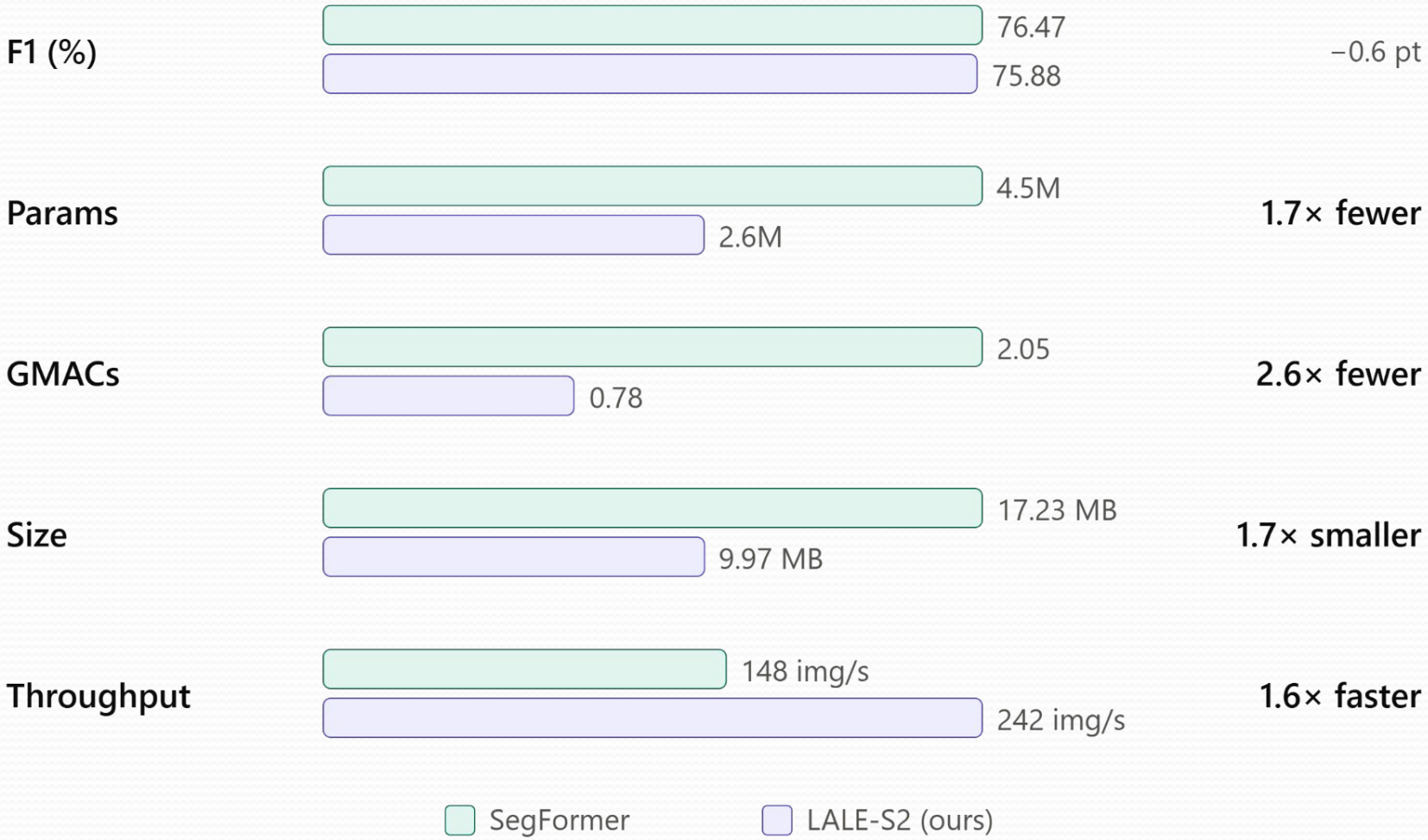


Results (ARAS)



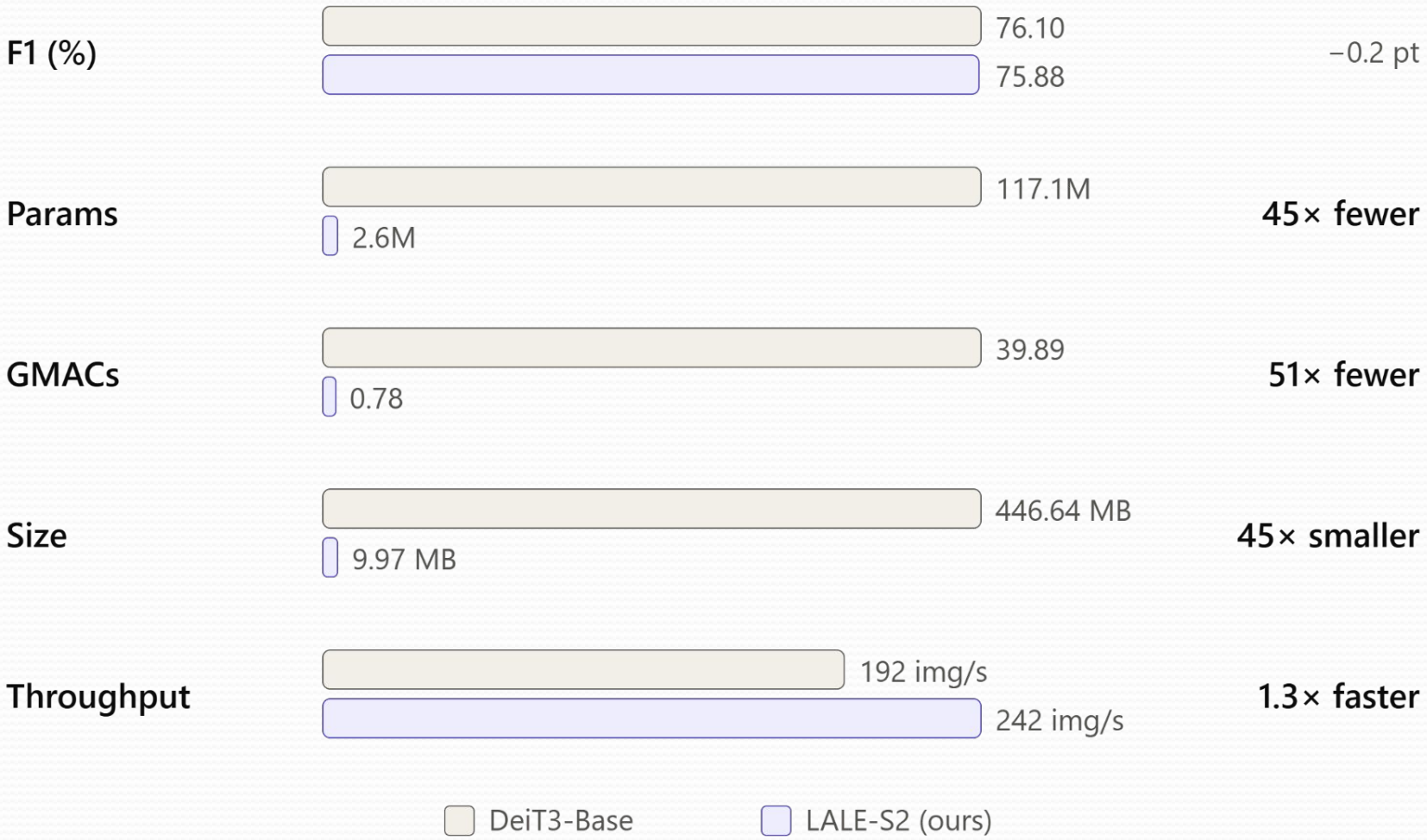


Results (ARAS)



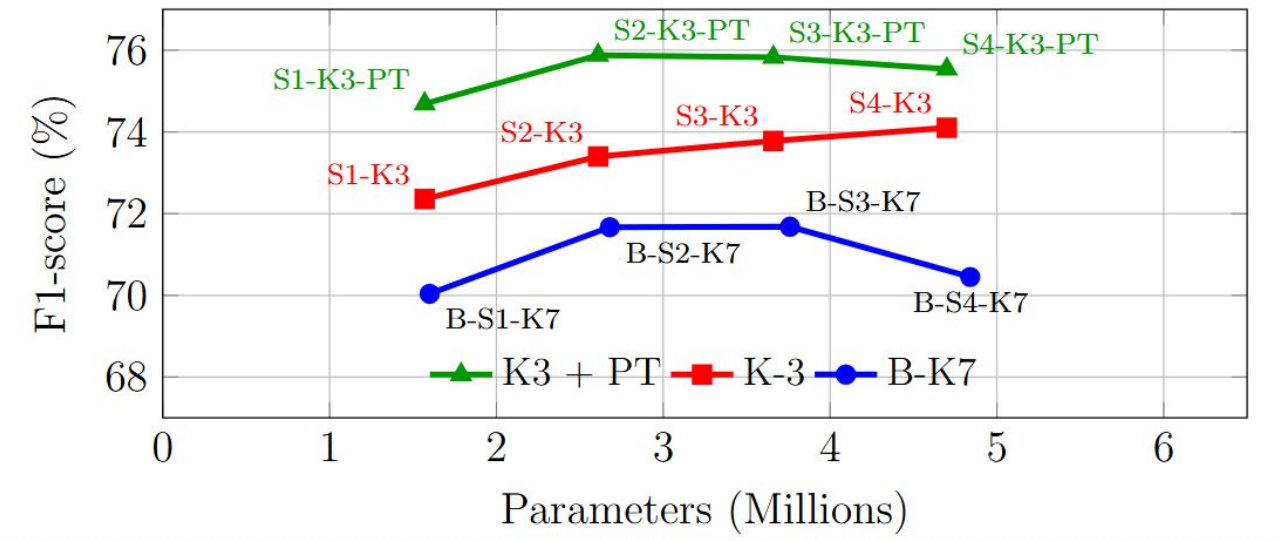
Throughput at batch size 1, FP32, 256×256 (1000 / latency)

Results (ARAS)



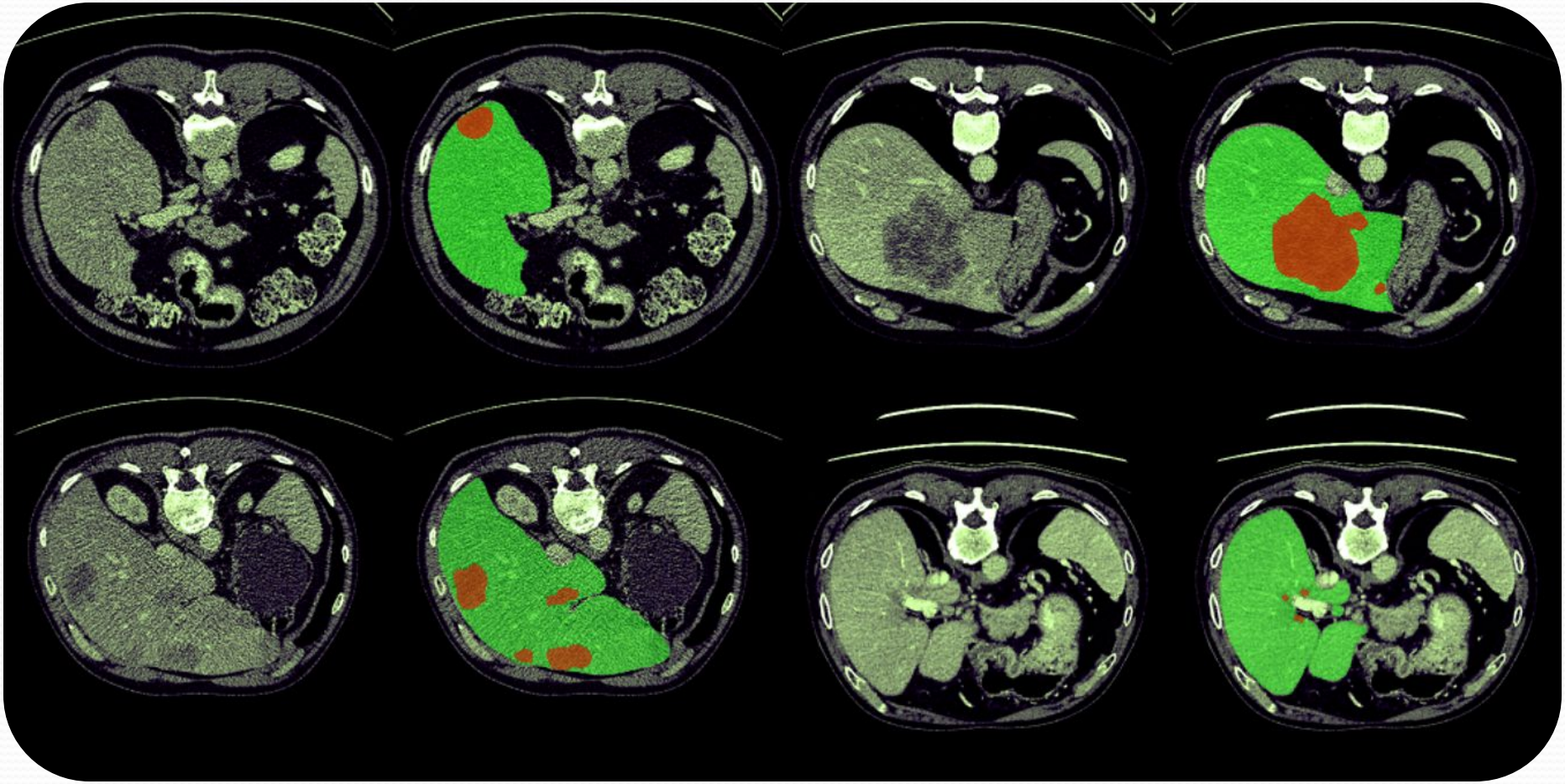
Ablation Results (ARAS)

Configuration	F1	Accuracy	Precision	Recall	IoU	Size (MB)	Throughput (img/s, bs=1)	Params (M)	GMACs
S4-K3-PT	75.54	83.82	76.20	74.93	63.27	17.94	174	4.7	1.16
S3-K3-PT	75.83	84.17	76.41	75.30	63.63	13.95	212	3.7	0.97
S2-K3-PT	75.88	84.12	76.39	75.42	63.67	9.97	242	2.6	0.78
S1-K3-PT	74.69	83.16	75.18	74.26	62.25	5.99	363	1.6	0.59
S4-K3	74.10	82.70	74.65	73.60	61.61	17.94	148	4.7	1.16
S3-K3	73.78	82.31	73.94	73.70	61.23	13.95	223	3.7	0.97
S2-K3	73.40	82.13	73.83	73.05	60.79	9.97	238	2.6	0.78
S1-K3	72.36	81.15	72.77	72.08	59.56	5.99	360	1.6	0.59
B-S4-K7	70.45	79.76	70.98	70.09	57.46	18.49	256	4.8	1.48
B-S3-K7	71.68	80.80	72.43	71.03	58.83	14.37	266	3.8	1.21
B-S2-K7	71.67	80.69	72.05	71.38	58.77	10.24	349	2.7	0.94
B-S1-K7	70.04	79.15	70.62	69.63	56.93	6.12	534	1.6	0.67



S: Scale B: Baseline K: Kernel PT: Pre-trained

Dataset: Liver and Tumor Segmentation (LiTS)

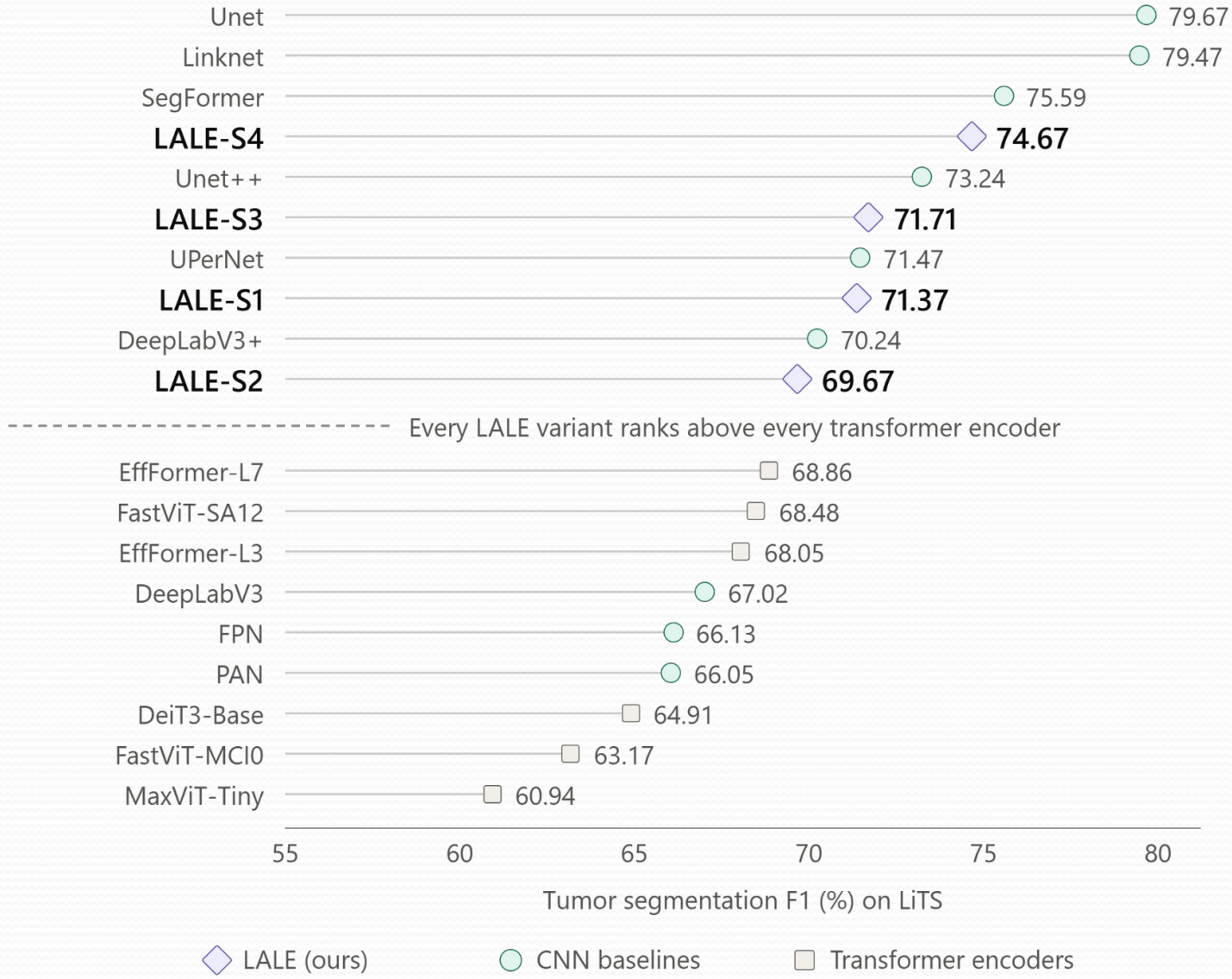


CT imagery slices, Green: Liver Red: Tumor

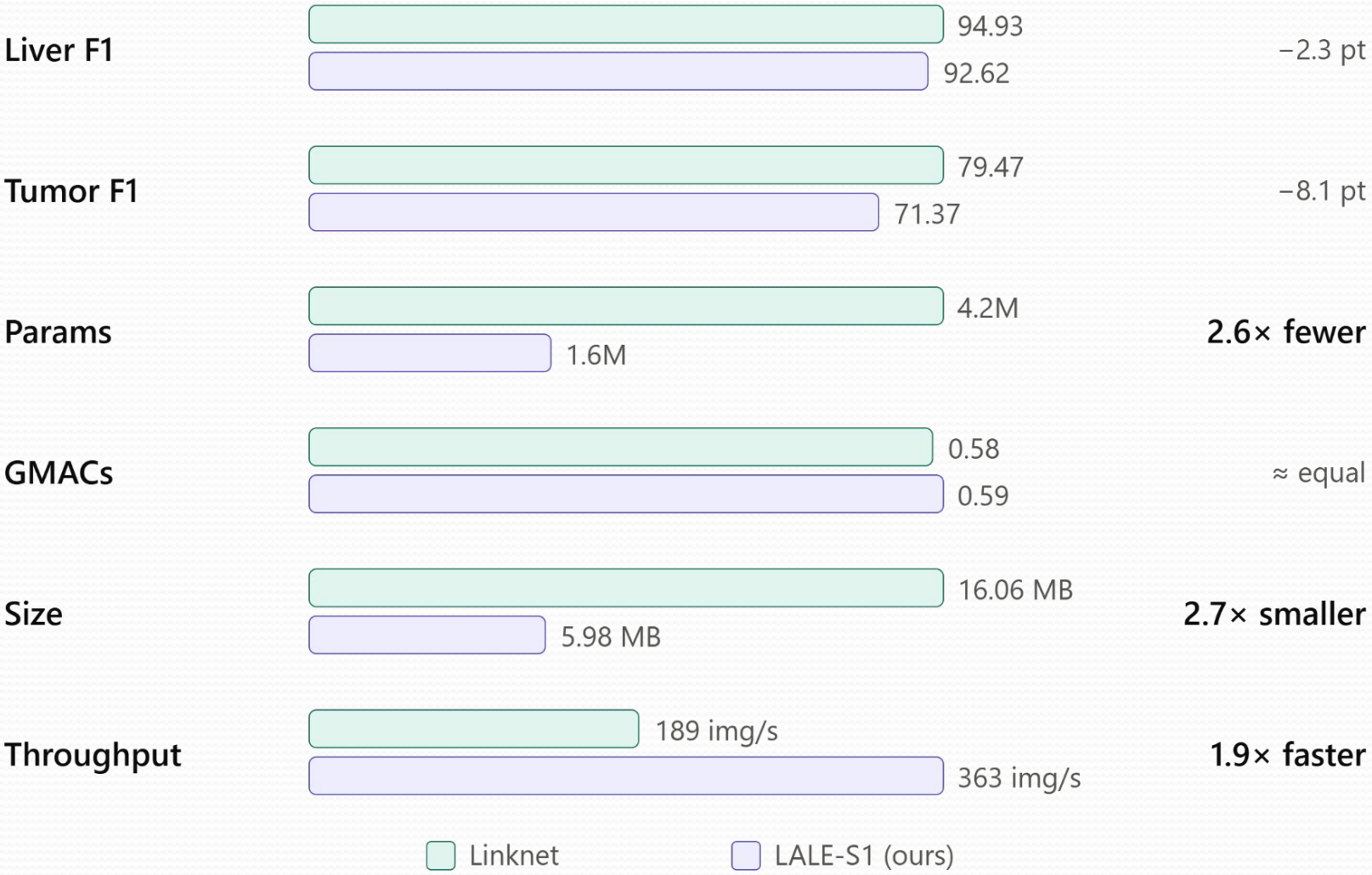
Results (LiTS)

Architecture	Liver Segmentation				Tumor Segmentation			
	F1	Precision	Recall	IoU	F1	Precision	Recall	IoU
DeepLabV3	94.68	93.91	95.46	88.52	67.02	71.39	63.15	50.40
DeepLabV3+	94.82	94.03	95.62	89.45	70.24	77.44	64.26	54.13
FPN	95.14	94.07	96.23	89.78	66.13	68.64	63.80	49.40
Linknet	94.93	93.64	96.25	89.21	79.47	82.18	76.94	65.94
PAN	93.83	91.76	95.99	87.46	66.05	73.47	59.99	49.31
Unet	95.06	93.64	96.53	90.59	79.67	82.30	77.20	66.21
UnetPlusPlus	95.43	94.11	96.79	90.30	73.24	83.59	65.18	57.78
UPerNet	94.69	93.95	95.43	89.20	71.47	79.18	65.13	55.61
Segformer	94.69	93.11	96.32	89.46	75.59	86.55	67.09	60.75
LALE-S1	92.62	91.86	93.38	85.55	71.37	72.13	70.62	55.49
LALE-S2	93.62	93.54	93.70	88.00	69.67	65.92	73.87	53.45
LALE-S3	94.13	94.09	94.17	87.92	71.71	75.27	68.47	55.90
LALE-S4	93.98	94.78	93.19	88.03	74.67	82.86	67.95	59.58
EffFormer-L3	93.40	91.63	95.24	87.52	68.05	70.67	65.61	51.57
EffFormer-L7	93.93	92.09	95.83	87.82	68.86	81.76	59.47	52.51
DeiT3-Base	95.13	94.30	95.97	89.40	64.91	83.06	53.27	48.05
MaxViT-Tiny	94.98	93.86	96.14	90.00	60.94	66.20	56.45	43.82
FastViT-SA12	92.73	91.41	94.09	86.44	68.48	73.18	64.34	52.06
FastViT-MCI0	95.01	93.86	96.19	89.90	63.17	64.62	61.78	46.16

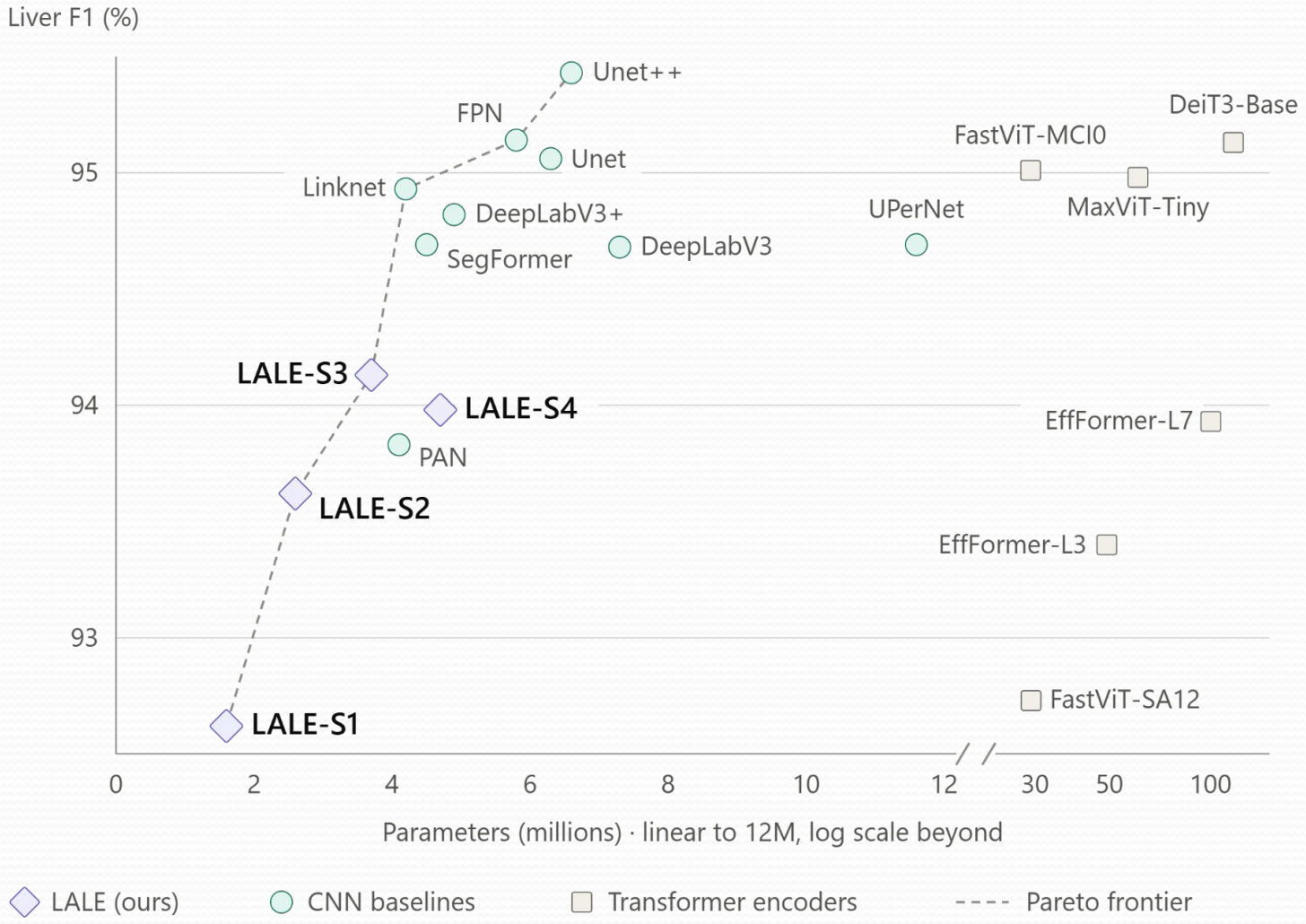
Results (LiTS)



Results (LiTS)



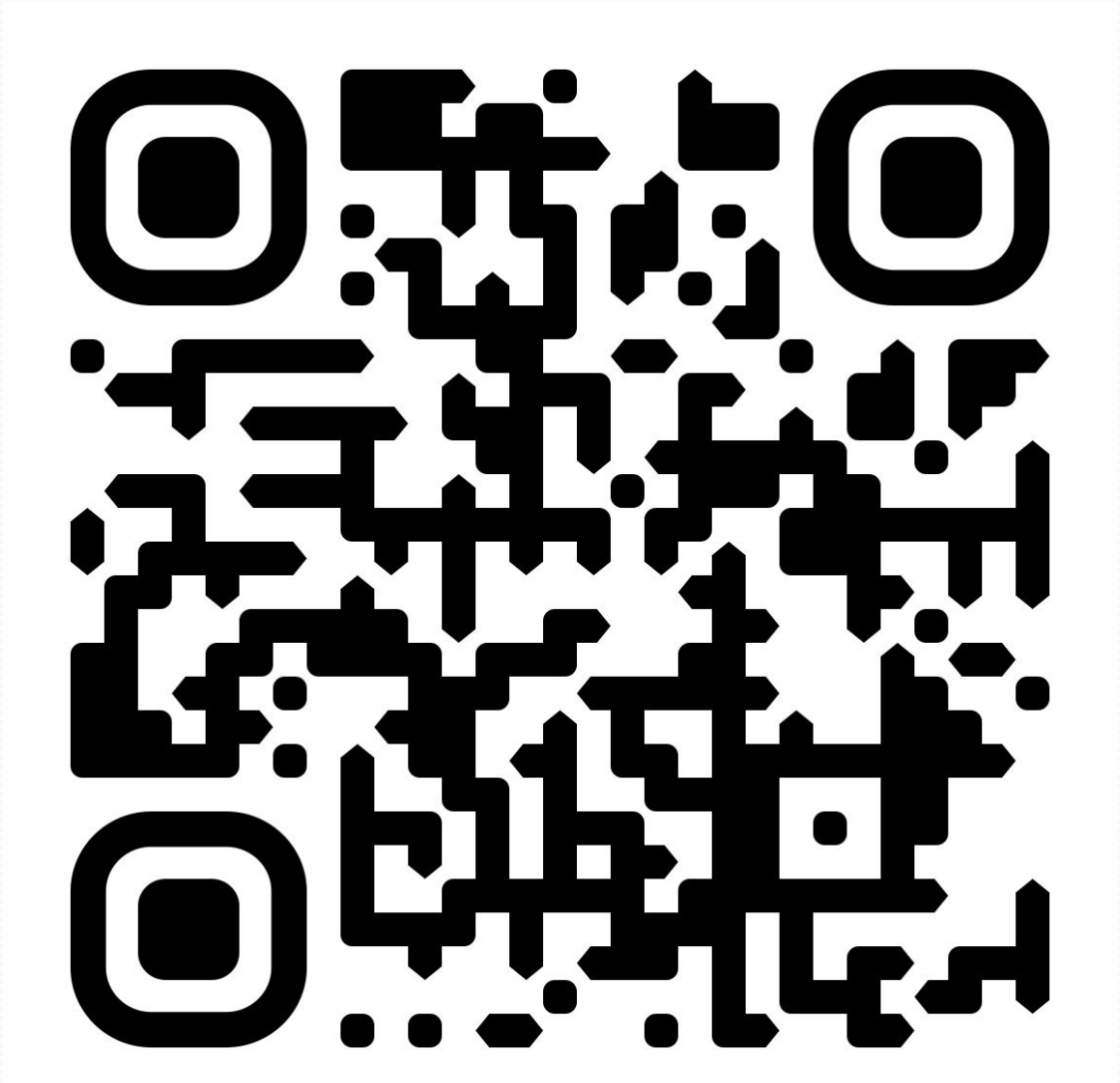
Results (LiTS)



Conclusion

- **Efficient** and **competitive** architecture
- For remote sensing **semantic segmentation**
- Works with **other domains** (medical imaging)
- **No external backbones**
- **End-to-end**, dense prediction
- **Limitations:** 1-2.6 % F1 drop, no edge evaluation
- **Future works:** Further ablation, edge deployment

Project Page



arxiv.org/abs/2606.02092 caglarmert.github.io/LALE github.com/caglarmert/LALE